Flight Test and Evaluation Report — Su-30MKI

AIRCRAFT AND SYSTEM TESTING ESTABLISHMENT

FLIGHT TEST AND EVALUATION REPORT

Su-30MKI — General Flight Characteristics and Level-Flight Evaluation

Report No.: FTER/TRG/SU30MKI/001 (Illustrative)

Aircraft: Sukhoi Su-30MKI

Test category: General flight-characteristics study

Document status: Training example — not an approved flight-test report

Test date / sortie / aircraft serial: To be entered from authorized records

Document control

Item	Details
Prepared by	Flight Test Evaluation Team — template
Reviewed by	Designated technical authority
Approved by	Authorized approving officer
Classification	To be assigned under applicable policy
Reference basis	Public aircraft information and general flight-test engineering principles

1. Summary

The Su-30MKI is a twin-seat, twin-engine, multirole fighter aircraft developed for Indian requirements and manufactured in India by Hindustan Aeronautics Limited (HAL). Public descriptions identify its thrust-vectoring engines, digital flight-control architecture, and integrated avionics as significant features.

hal-india.co.in

+2

This report establishes a documentation framework for studying general flight characteristics, with particular emphasis on steady level flight. The evaluation framework covers aircraft configuration, test conditions, data quality, longitudinal stability, trim behaviour, engine indications, flight-control response, and repeatability.

No actual sortie data or verified Su-30MKI test measurements have been supplied. Consequently, this report does not claim that a flight was conducted, that any acceptance criterion was met, or that the aircraft was cleared for any operation.

2. Aim

To establish a structured framework for documenting and evaluating the general flight characteristics of the Su-30MKI during a representative steady-flight condition, using authorized aircraft documentation and verified flight-test data.

3. Scope

The report covers:

- Aircraft identification and configuration documentation.
- Environmental and instrumentation data requirements.
- General principles for assessing steady-flight behaviour.
- Data validation, uncertainty, and reporting requirements.
- Limitations, conclusions, and follow-up documentation.

It does not establish aircraft-specific operating procedures, performance limits, test-card parameters, or flight-clearance decisions.

4. Aircraft description

8

The following summary is based on public descriptions and is intended for general technical context.

System / feature	General description
Aircraft type	Twin-seat, twin-engine multirole fighter
Propulsion	Two AL-31FP turbofan engines
Flight controls	Digital fly-by-wire control architecture
Manoeuvring feature	Thrust-vectoring capability
Avionics	Integrated navigation, communications, and mission systems
Intended roles	Air superiority and multiple other military aviation roles

+2

5. Test objectives

The following are general engineering objectives for a training report, not a flight authorization or a prescribed test sequence.

1. Establish the identity and configuration of the aircraft represented by the data.
2. Verify the completeness and consistency of recorded flight parameters.
3. Assess whether the available data support a steady-flight condition.
4. Examine the consistency of recorded air-data, attitude, engine, and flight-control indications.
5. Document data uncertainty, anomalies, and any limitations that prevent a defensible conclusion.

6. Conditions relevant to the evaluation

6.1 Aircraft configuration

The following information should be completed from authorized records before any real test data are analysed.

Parameter	Required entry
Aircraft serial number	Authorized record
Aircraft modification standard	Verified configuration
Engine configuration	Verified maintenance record
Aircraft all-up weight	Recorded value and units

Centre of gravity	Authorized loading documentation
External configuration	Recorded configuration
Flight-control system status	Recorded technical status
Maintenance status	Relevant serviceability records
Instrumentation configuration	Installed and calibrated equipment

6.2 Environmental conditions

Environmental information is necessary to interpret flight data correctly and compare observations collected on different occasions.

Parameter	Data to record
Date and time	Timestamp and time reference
Atmospheric conditions	Recorded temperature and pressure
Wind	Available meteorological observations
Atmospheric density	Calculated from validated inputs
Turbulence	Qualitative assessment or recorded measurement
Airfield and test area	Authorized documentation
Instrument calibration	Calibration status and validity

6.3 Instrumentation and data quality

The evaluation should identify the source, units, resolution, sampling characteristics, calibration status, and uncertainty of each parameter used in the analysis.

Typical general-purpose flight-test data categories include:

- Air-data measurements and derived quantities.
- Aircraft attitude and angular-motion measurements.
- Engine monitoring indications.
- Flight-control and trim indications.
- Navigation and time-synchronization data.
- Environmental and configuration records.

The actual instrumentation suite must be established from the aircraft's authorized test documentation. No specific instrumentation fit is assumed here.

7. Test methodology — steady level flight

7.1 Definition

For this report, steady level flight refers to a condition in which aircraft altitude and flight state remain approximately constant over a defined observation interval, within documented measurement uncertainty and the applicable test criteria.

A simplified conceptual relationship is:

\\[L \approx W \\]

where:

- \\(L\\) = aerodynamic lift.
- \\(W\\) = aircraft weight.

For a simplified, steady, unaccelerated condition, lift approximately balances weight. Actual aircraft behaviour depends on the full force and moment balance, atmospheric conditions, aircraft configuration, and control-system response.

7.2 Evaluation sequence

Stage 1 — Establish data validity

Confirm aircraft identity, configuration records, time synchronization, instrumentation status, and data completeness.

Stage 2 — Review recorded parameters

Inspect the available altitude, air-data, attitude, engine, and control-system records for gaps, inconsistent units, and anomalous readings.

Stage 3 — Analyse the observation interval

Evaluate trends, variability, sensor agreement, and measurement uncertainty using the approved test criteria.

Stage 4 — Document the findings

Separate verified observations from interpretations, identify unresolved anomalies, and record whether the available evidence is sufficient for a conclusion.

8. Analysis and results

Status: Pending validated flight-test data. The following tables are reporting templates; they contain no measured Su-30MKI results.

8.1 General flight-state assessment

Assessment item	Evidence required	Result
Altitude stability	Time history and measurement uncertainty	Pending
Air-data consistency	Validated sensor records	Pending
Attitude stability	Recorded attitude data	Pending
Longitudinal behaviour	Relevant flight-data time histories	Pending
Engine indication consistency	Recorded engine-monitoring data	Pending
Control-system indication consistency	Relevant recorded parameters	Pending
Repeatability	Comparable validated observations	Pending

8.2 Performance data recording sheet

Parameter	Unit	Recorded value
Pressure altitude	ft or m	Not supplied
Indicated airspeed	kt or km/h	Not supplied
Calibrated airspeed, if available	kt or km/h	Not supplied
Mach number, if available	Dimensionless	Not supplied
Outside air temperature	°C	Not supplied
Aircraft mass	kg	Not supplied
Engine indications	As defined in source	Not supplied
Observation interval	s or min	Not supplied
Measurement uncertainty	Parameter-specific	Not supplied

The units, definitions, and sources must be retained consistently when data are imported into the final report.

8.3 Data interpretation

The analysis should distinguish between three categories:

1. Observation: What the recorded data directly show.

2. Interpretation: What the observations suggest under the stated assumptions.
3. Conclusion: What can be supported after uncertainty, data quality, and acceptance criteria are considered.

For example, a relatively stable altitude trace alone does not establish that all aspects of the aircraft's flight state were steady. Sensor quality, the observation interval, other flight parameters, and the applicable criteria must also be considered.

8.4 Graphs for the final report

Once validated data are available, the report may include the following figures:

- Altitude versus time.
- Airspeed versus time.
- Aircraft attitude versus time.
- Engine indications versus time.
- Relevant control-system indications versus time.
- Measurement uncertainty or variability plots.

Each figure should identify its source, units, observation interval, and any data-processing method. Illustrative graphs must be explicitly labelled as simulated.

9. Limitations and error considerations

The following issues can affect the reliability of a flight-test assessment:

Potential limitation	Effect on interpretation
Sensor calibration uncertainty	Limits confidence in absolute measurements
Time synchronization errors	Can distort relationships between parameters
Missing or corrupted data	May prevent complete assessment
Atmospheric variability	Can influence comparisons between observations
Configuration differences	Can make separate observations non-comparable
Processing or rounding	Can introduce apparent differences or conceal variation
Incomplete acceptance criteria	Prevents a formal pass/fail conclusion

An uncertainty budget should be prepared using the actual measurement system and applicable engineering procedures. No numerical uncertainty is assigned here because the necessary data are unavailable.

10. Findings

Based on the information currently available, the report establishes the scope and documentation structure for a general flight-characteristics evaluation of the Su-30MKI.

The aircraft's broad configuration is supported by public references. However, aircraft-specific flight behaviour during a particular sortie cannot be established from the general description alone.

The following items remain outstanding:

- Aircraft and configuration records.
- Actual flight-test data.
- Instrument calibration and uncertainty information.
- Approved acceptance criteria.
- Review and approval by the designated technical authority.

11. Conclusion

This report provides a structured framework for documenting the general flight characteristics of the Su-30MKI, with a focus on steady level flight.

No formal assessment of flight performance, compliance, airworthiness, or operational suitability can be made from the information presently available. Such conclusions require validated data, applicable approved criteria, and review by appropriately authorized personnel.

12. Recommendations

1. Maintain traceability between each reported result and its original data source.
2. Record aircraft configuration and environmental conditions alongside the data.
3. Document measurement uncertainty and unresolved anomalies.
4. Clearly distinguish illustrative examples from actual test observations.
5. Obtain authorized technical review before issuing a report as an official FTER.

13. Appendices

- Appendix A: Aircraft and configuration record.
- Appendix B: Flight-data source register.
- Appendix C: Instrumentation and calibration register.
- Appendix D: Data-quality and uncertainty assessment.
- Appendix E: Figures, plots, and supporting observations.
- Appendix F: Review comments and approval record.

References

1. Hindustan Aeronautics Limited — official [Su-30MKI aircraft overview](https://hal-india.co.in/product/su-30-mki?utm_source=chatgpt.com).

   [image](https://www.google.com/s2/favicons?domain=https://hal-india.co.in\&sz=32)

   hal-india.co.in

2. Government of India, Ministry of Defence — [Annual Report 2023–24](https://www.ddpmod.gov.in/sites/default/files/57cfea92938b03f670763d972acabf946f5cf7bd47c6252b78aede2f466d0214/aa2dde469a046132ce341e7e686e18eba17f1a007e9519a8658715109a3a729a.pdf?utm_source=chatgpt.com).

If you want this report as a properly formatted Word or PDF document, I can prepare a document with a cover page, table of contents, numbered sections, editable result tables, figure placeholders, and signature blocks.